



Deep Learning (Master's degree in AI)

Lab assignment 3 - Latent Space Classification (2025-2026)

INSTRUCTIONS:

- **Deadline:** June 05, 23:59.

- **Objectives**

The goal of this lab assignment is to investigate the use of an Autoencoder as a dimensionality reducer prior to a classifier.

- **Dataset**

We are going to use the STL-10 dataset. STL-10 contains 13,000 labeled images of 96×96 pixels in color (5,000 for training and 8,000 for testing). It also includes 100,000 additional untagged images for unsupervised learning (see practice 1 for more information).

The images belong to 10 possible categories (airplane, bird, car, cat, deer, dog, horse, monkey, ship, truck) and are perfectly distributed (500 training images and 800 test images per class).

In this assignment, we use the unlabelled portion of the dataset to train an auto-encoder in an unsupervised manner, allowing the model to learn a compact and meaningful representation of the images. Once trained, the encoder part of the autoencoder is retained and used as a dimensionality reduction module. Finally, a classifier is trained on top of the latent representations produced by the encoder using the labelled training images, and its performance is evaluated on the test set.

- **Tasks**

1. **Preliminary Research:** begin by reviewing the fundamental concepts of Autoencoders (AEs), including their architecture, training process, and application as generative and representation learning models.

Review Practice 1 and analyze the performance of the different supervised models you trained. The reference performance baseline for this practice will be the one obtained using dense (fully connected) models.

2. **Unsupervised Training:** using the unlabelled split of the STL-10 dataset, design and train a convolutional autoencoder. The goal is to learn a compact latent representation of the input images through unsupervised learning. Justify the main architectural choices of the encoder and decoder.

3. **Latent Space Classification:** using the labelled training set, train a classifier on top of the frozen encoder by operating exclusively in the latent space. Compare its performance against the baseline dense models obtained in Practice 1. Apply appropriate regularization techniques and discuss the impact on overfitting and generalization.
4. **Fine-Tuning and Regularization (Optional):** experiment with partial or full fine-tuning of the encoder using the classification loss.

Students may use AI-based tools (e.g., large language models) as a support resource for exploring neural network architectures, understanding design alternatives, and clarifying technical concepts, provided they understand and can justify all architectural decisions made. Blindly copying AI-generated code or architectures without comprehension is not acceptable.

Training the autoencoder on the full unlabelled portion of the STL-10 dataset can be computationally demanding. To accommodate hardware limitations, students may train the autoencoder using a subset of the unlabelled data, provided that the subset is sufficiently large to learn meaningful representations. The choice of subset size should be justified.

■ Practice groups

- The practice will preferably be done in pairs (it can be also done alone) and both members of the group will be responsible and should know everything that is delivered on their behalf.
- No change of practice group will be made during the course. The group can be undone and in this case its members will continue to carry out the practices alone.

■ Submission

- The exercises will be developed using Jupyter Notebooks. If you use more than one put them together in a ZIP file for submission.
- Be organized when naming notebooks (use numbers to describe order) and notebook sections (using Markdown headings).
- **Each notebook should include:**
 - The first cell of each notebook must be the full names of the authors.
 - The notebook will be saved with the results of its execution included.
 - The code should be commented and shall be accompanied by jupyter text cells with an explanatory report containing a description of the process, detailing the encountered problems and justifying your decisions.
 - AI tools using declaration: see evaluation criteria
- **Submission process**
 - The exercises will be submitted using the virtual campus of each university:
 - ◊ Universidade da Coruña: <https://campusvirtual.udc.gal/>
 - ◊ Universidade de Vigo: <https://moovi.uvigo.gal/>

- ◊ Universidade de Santiago de Compostela: <https://cv.usc.es/>
- Each member of the practice group must submit the notebook in their corresponding Moodle task.
- There is a strict deadline for each assignment. Past due submissions will be rejected.

■ Evaluation criteria

- Accuracy of the models: evaluation and comparison of performance of the models against performance obtained in practice 1.
- Quality of the design
 - Appropriate data preprocessing and training techniques
 - The network design follows the recommendations on how to create them.
 - The regularization techniques are correctly applied and a thorough analysis has been made to see which ones work best.
- The process is sufficiently detailed and the decisions are justified.
- The results are commented and interpreted correctly.
- All the results are compared and summarized.
- AI Tool Usage Declaration: include a markdown cell at the end of your notebooks commenting if you used any AI tools (e.g., ChatGPT, GitHub Copilot, etc.) during the completion of this lab. Describe briefly how you used them and how useful they were for you. This is required for academic integrity: using AI tools will not affect your grade but declaring that you used them (or not) will.